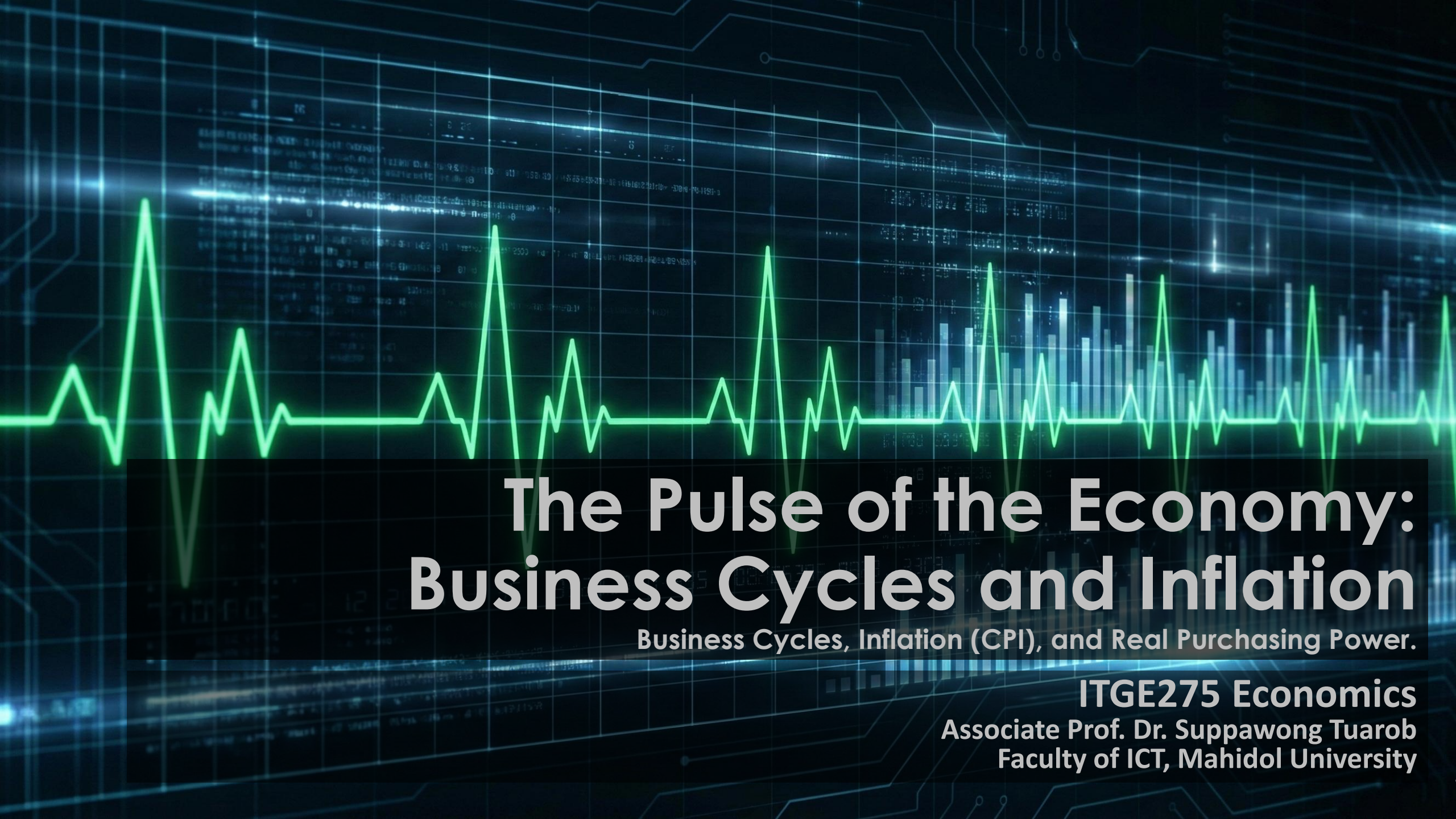


The Pulse of the Economy: Business Cycles and Inflation

Business Cycles, Inflation (CPI), and Real Purchasing Power.

ITGE275 Economics

Associate Prof. Dr. Suppawong Tuarob

Faculty of ICT, Mahidol University

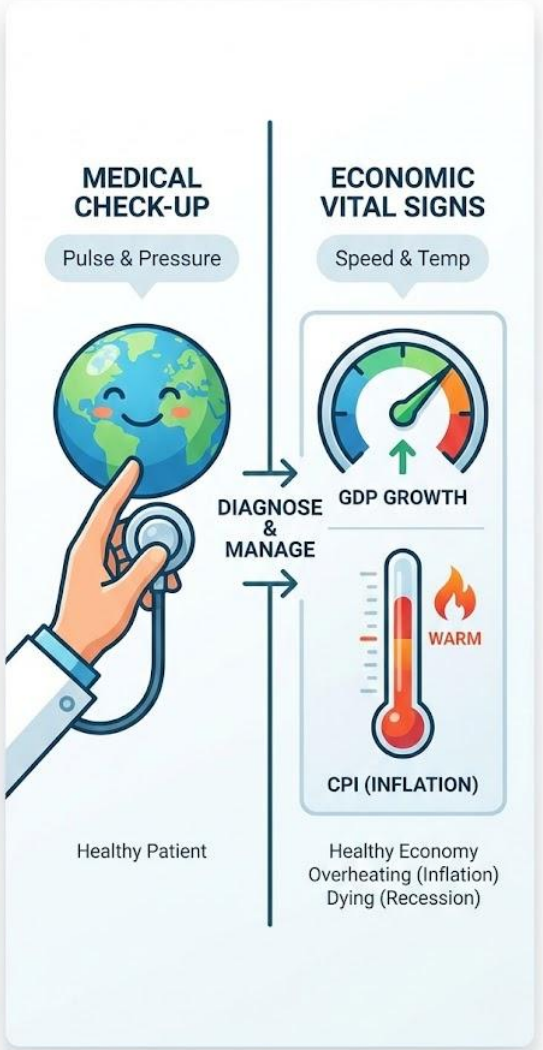

ดัชนีราคาผู้บริโภคของไทย
เดือนมกราคม 2569
เท่ากับ 99.91
อัตราเงินเฟ้อทั่วไป
ลดลงที่ร้อยละ 0.66 (YoY)
กระทรวงพาณิชย์
Ministry of Commerce
TPSO
สำนักงานนโยบายและยุทธศาสตร์การค้า
Trade Policy and Strategy Office


MONEY & BANKING
พาณิชย์ยันยังไม่เงินฝืด!
เงินเฟ้อไทย ม.ค.69 ติดลบ 0.66%
ต่อเนื่องเดือนที่ 10 คาด Q2/69 พลิกบวก
5 ก.พ. 2569


ประชาชาติ
กนง.ประกาศเป้าเงินเฟ้อปี'69
รักษาเสถียรภาพราคา
ยืดหยุ่นรับความผันผวนได้
6 ม.ค. 2569

Economic Vital Signs

- **Concept:** Just as a doctor measures pulse and blood pressure to judge a patient's health, economists measure **Business Cycles** and **Inflation**.
- **The Goal:** To differentiate between a healthy growing economy and one that is "overheating" (High Inflation) or "dying" (Recession/Deflation).
- **Key Metrics:**
  - **GDP Growth:** The speed of the engine.
  - **CPI (Inflation):** The temperature of the engine.



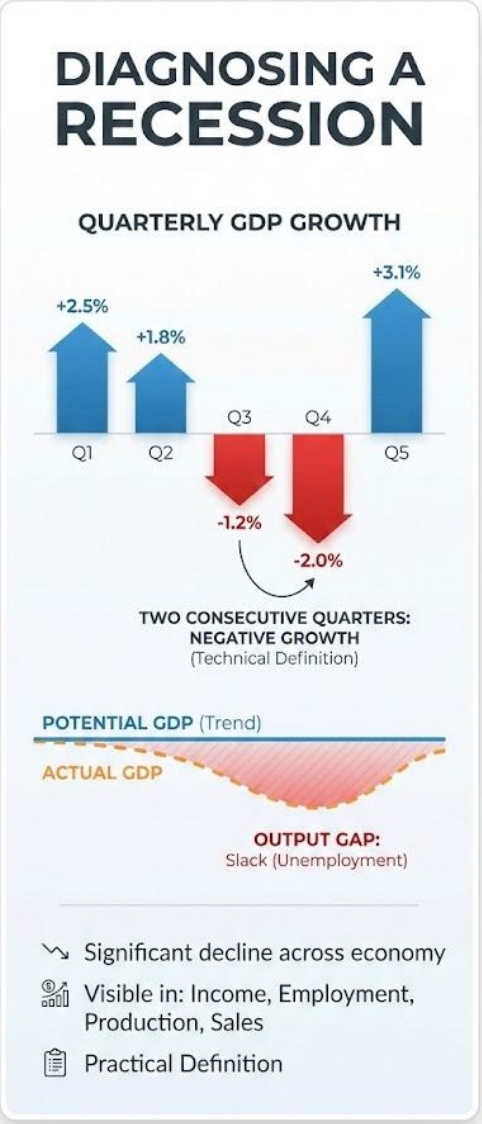
The Business Cycle Defined

- **Definition:** The natural, recurring fluctuation of economic activity around a long-term growth trend. Economies do not grow in a straight line; they oscillate.
- **The 4 Phases:**
  - **Expansion:** Growth, rising employment, rising prices.
  - **Peak:** The height of the boom; maximum output.
  - **Contraction (Recession):** Falling GDP, rising unemployment.
  - **Trough:** The bottom of the crash before recovery begins.



Diagnosing a Recession

- **Technical Definition:** Two consecutive quarters (6 months) of negative Real GDP growth.
- **Practical Definition:** A significant decline in economic activity spread across the economy (visible in real GDP, real income, employment, industrial production, and wholesale-retail sales).
- **The "Output Gap":** When actual GDP falls below potential GDP (the Trend), creating slack in the system (unemployment/idle factories).



Case Study: 1997 Tom Yam Kung Crisis



- **Context:** Thailand, 1997. A classic **Financial/Currency Crisis**.
- **Trigger:** The Thai Baht was unpegged from the US Dollar and collapsed. From 25 → 50 ฿/$.
- **Mechanism:**
  - Currency devaluation → Import prices skyrocketed.
  - Foreign debt doubled in local currency terms.
  - Result: Massive bankruptcies and high **Cost-Push Inflation**.



The 1997 "Tom Yam Kung Crisis" was a severe financial collapse in Thailand, starting July 2, 1997, caused by over-borrowing, a real estate bubble, and unsustainable currency speculation. Thailand was forced to float the Thai Baht, causing its value to plummet by over 50%, forcing IMF intervention, bankrupting companies, and triggering a wider Asian financial crisis.

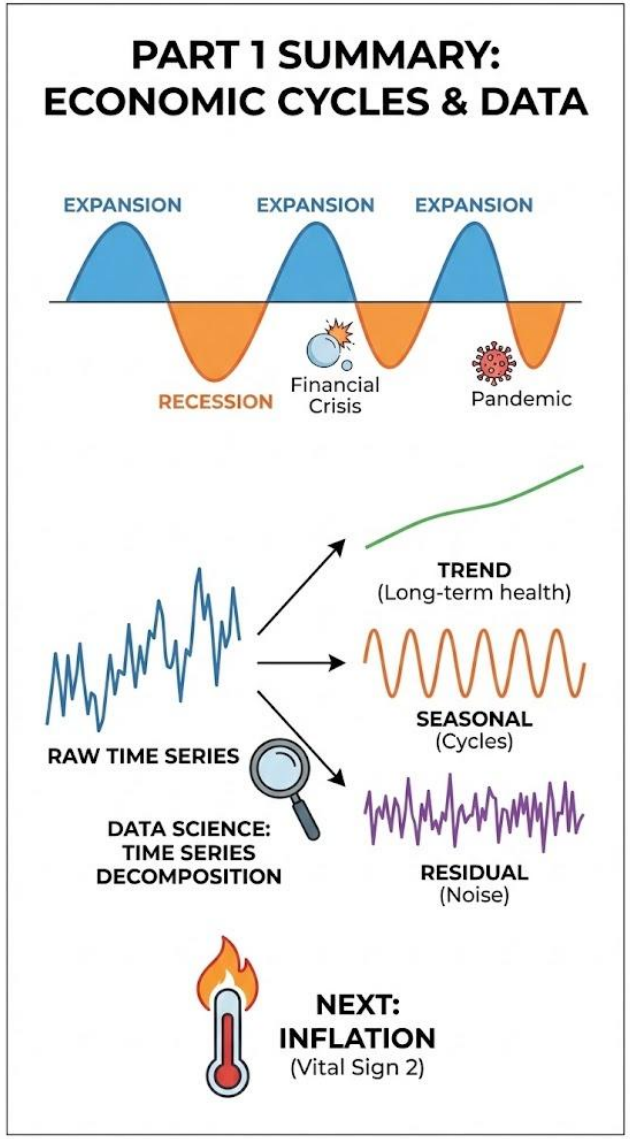
Case Study: 2020 COVID-19 Recession

- **Context:** Global, 2020. A unique **Exogenous Shock**.
- **Trigger:** Pandemic lockdowns and physical restrictions.
- **Mechanism:**
  - **Supply Shock:** Factories closed.
  - **Demand Shock:** Consumers stayed home (no travel, dining).
- **Result:** Unlike 1997, prices didn't spike immediately; they often fell (Deflation) due to collapsed demand.



Part 1 Summary

- Economies breathe in cycles (Expansion/Recession).
- Not all recessions are the same (Financial Crisis vs. Pandemic).
- **Data Science Role:** We use Time Series Decomposition to separate the "Trend" (Long-term health) from the "Cycle" (Short-term fluctuation).
- **Next:** The second vital sign—Inflation.



Part 2: Inflation & The CPI

What is Inflation?

- **Definition:** The rate at which the general level of prices for goods and services is rising, and consequently, the purchasing power of currency is falling.
- **Not just one price:** If *only* iPhones get expensive, that's not inflation. Inflation is when *everything* (Food, Rent, Gas) gets expensive together.
- **Deflation:** A general decline in prices (often dangerous as it signals weak demand).
- **Disinflation:** A slowing down of the inflation rate (e.g., from 5% to 3%).



The Consumer Price Index (CPI)

- **The Metric:** How do we measure the "Cost of Living"?
- **The Basket of Goods:** A representative collection of items the average citizen buys.
  - Food (Rice, Eggs).
  - Energy (Gasoline, Electricity).
  - Services (Rent, Netflix, Medical).
- **CPI Formula:**

$$CPI_t = \frac{\text{Cost of Basket}_t}{\text{Cost of Basket}_{base}} \times 100$$

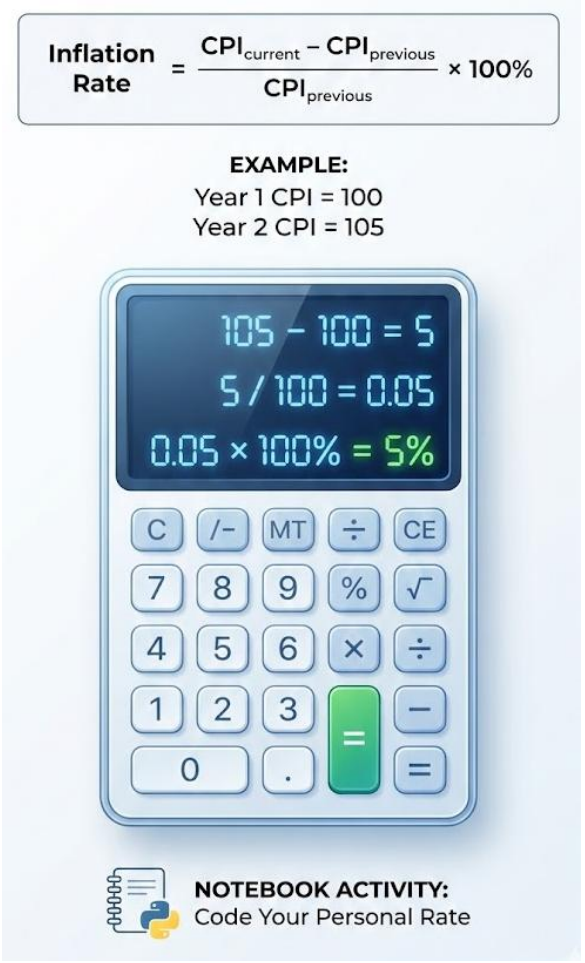


Calculating Inflation Rate

- Inflation is the *percentage change* in the CPI.
- **Formula:**

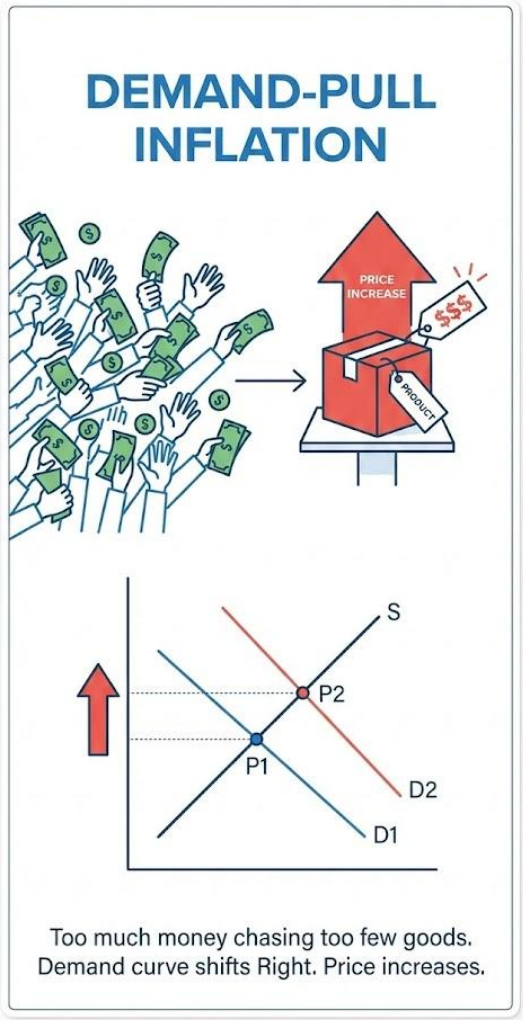
$$\text{Inflation Rate} = \frac{CPI_{current} - CPI_{previous}}{CPI_{previous}} \times 100\%$$

- **Example:**
  - Year 1 CPI = 100
  - Year 2 CPI = 105
  - Inflation = (105-100)/100 = 5%
- **Notebook Activity:** We will code a function to calculate your *personal* inflation rate based on your spending habits.



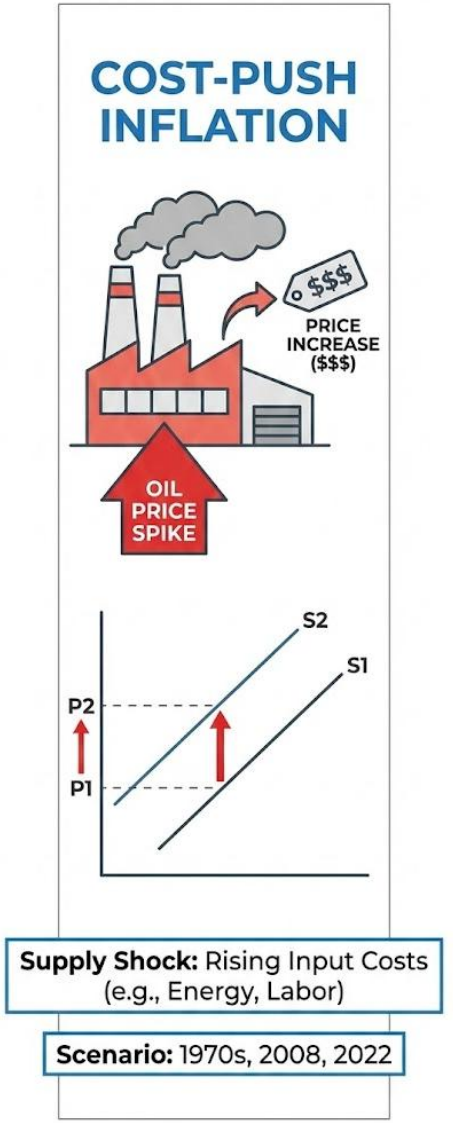
Types of Inflation: Demand-Pull

- "**Too much money chasing too few goods.**"
- **Cause:** When aggregate demand in an economy strongly outweighs aggregate supply.
- **Scenario:** The economy is booming, unemployment is low, everyone has cash, so they bid up prices.
- **Visual:** Demand curve shifts Right. Price increases.



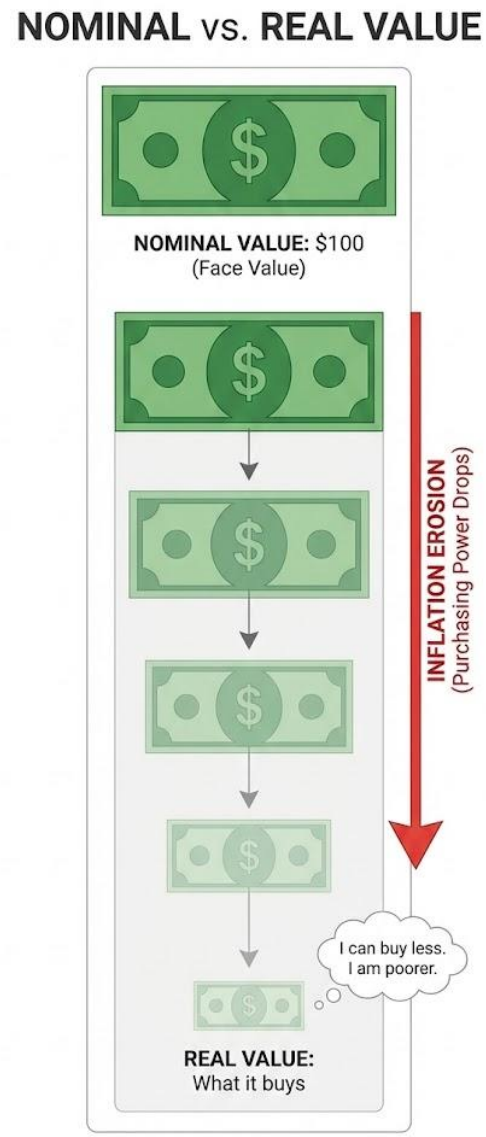
Types of Inflation: Cost-Push

- "**Supply shock**."
- **Cause:** Increase in the cost of production inputs (Labor, Raw Materials) creates inflation even if demand hasn't changed.
- **Scenario:** Oil prices spike (1970s, 2008, 2022). Since oil is used to transport *everything*, the price of *everything* goes up.
- **Visual:** Supply curve shifts Left. Price increases.



Nominal vs. Real Value

- **Nominal Value:** The face value of money (e.g., "I earned 1000 Baht").
- **Real Value:** What that money can actually buy (Purchasing Power).
- **The Erosion Effect:** If your income stays the same (Nominal) but Inflation is 5%, your Real Income has dropped by 5%. You are poorer.



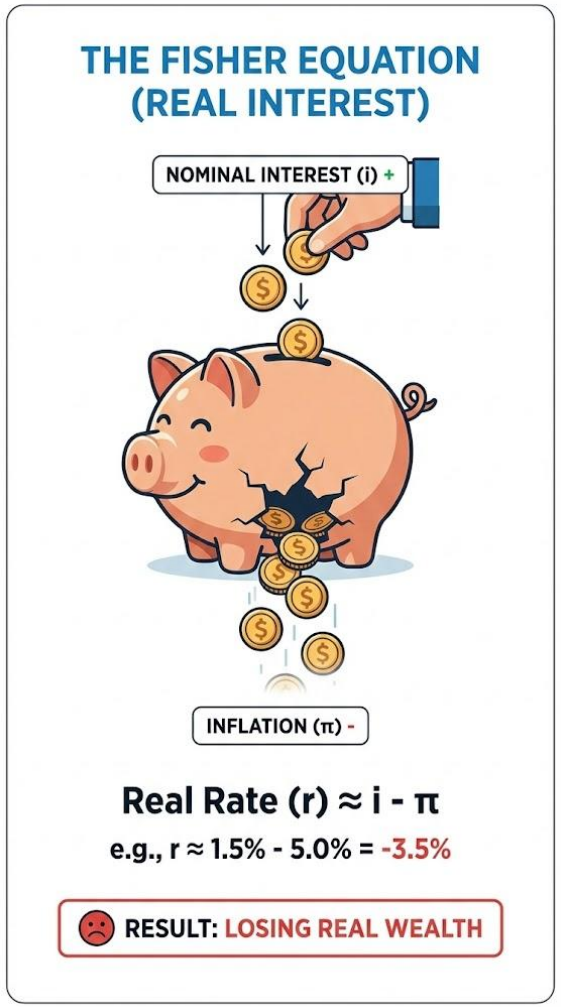
The Fisher Equation (Real Interest)

- How does inflation affect savings?
- **Nominal Interest Rate (i):** What the bank pays you (e.g., 1.5%).
- **Inflation Rate (π):** The erosion of value (e.g., 5.0%).
- **Real Interest Rate (r):**

$$r \approx i - \pi$$

$$r \approx 1.5\% - 5.0\% = -3.5\%$$

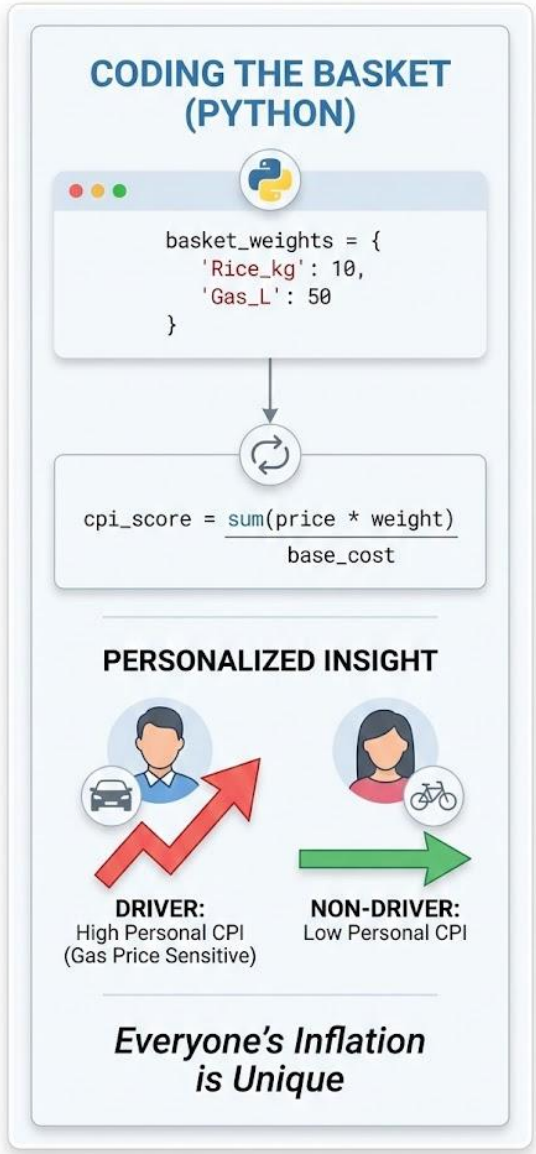
- **Conclusion:** If you save money in a low-interest account during high inflation, you are **losing** real wealth.



Part 3: Computational Analysis

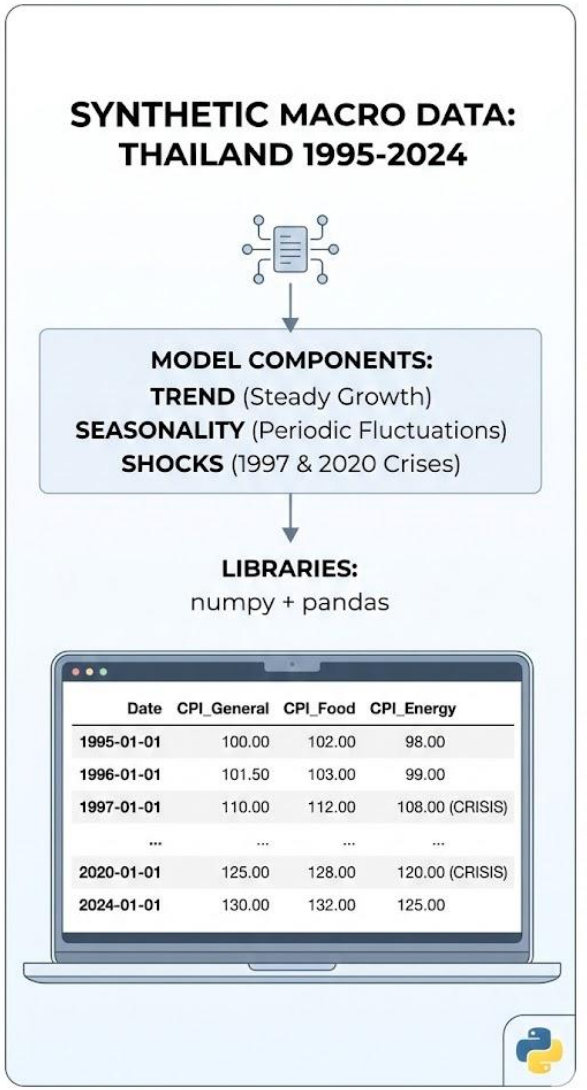
Coding the Basket of Goods

- In the tutorial, we define a Python dictionary to represent a user's lifestyle.
- **Weights:** Fixed quantities (e.g., Rice: 10kg, Gas: 50L).
- **Prices:** Variable inputs.
- **Logic:** We sum the Price * Weight for the Base Year and Current Year to derive the personal CPI.
- **Insight:** Everyone has a different personal inflation rate. If you don't drive, gas price spikes don't affect you as much.



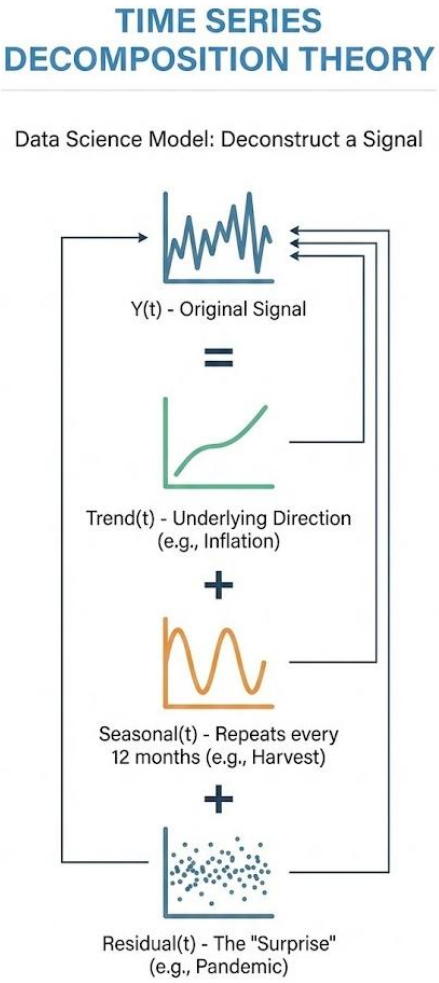
Generating Synthetic Macro Data

- To analyze history, we generate a dataset matching Thailand's economic timeline (1995-2024).
- **Components Modeled:**
  - **Trend:** Steady growth.
  - **Seasonality:** Periodic fluctuations.
  - **Shocks:** Manual injection of the 1997 and 2020 crises.
- **Library:** numpy and pandas for time series generation.



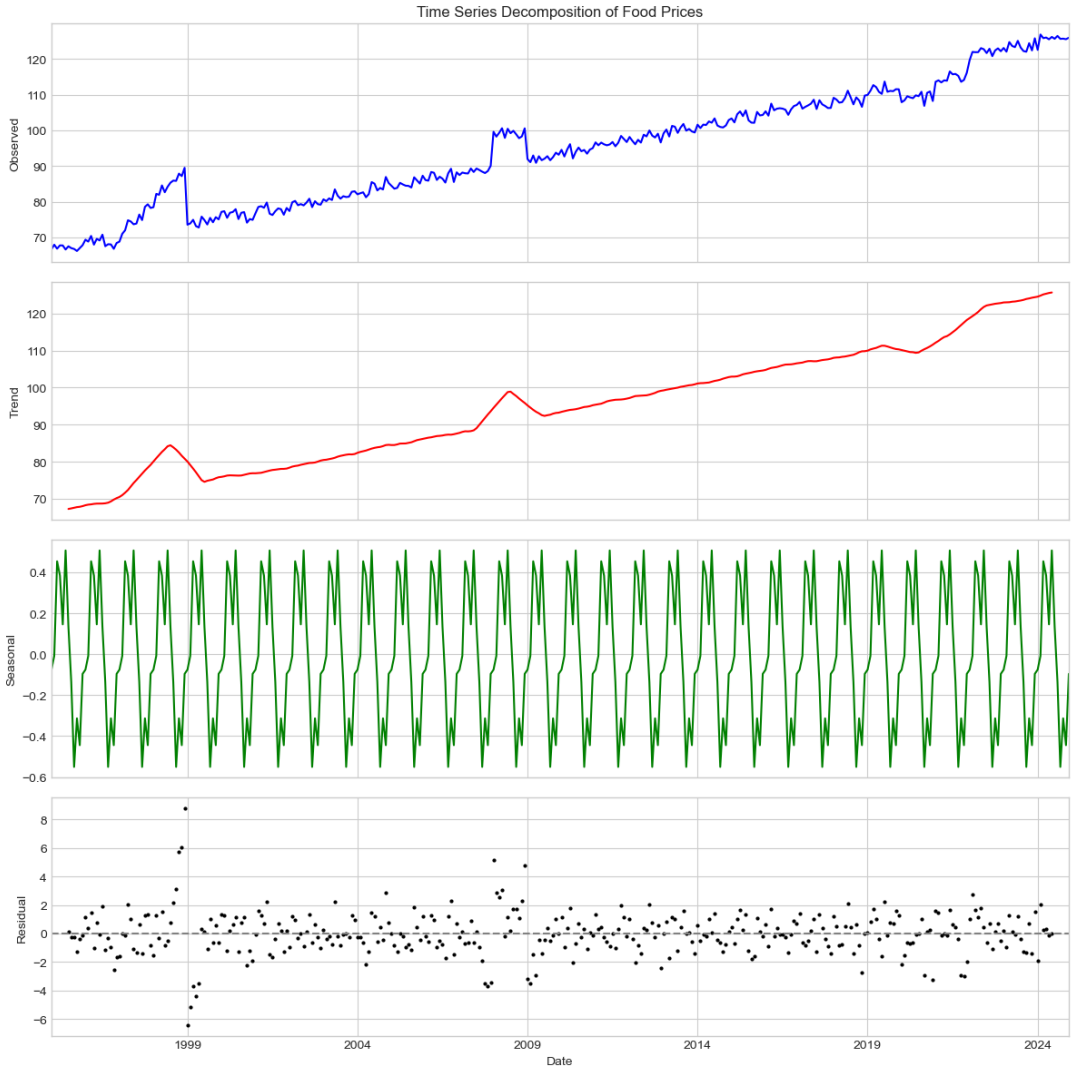
Time Series Decomposition Theory

- Data Science allows us to deconstruct a signal.
- **Model:** $Y(t) = Trend(t) + Seasonal(t) + Residual(t)$
- **Trend:** The underlying direction (Inflation is usually positive).
- **Seasonal:** Repeats every 12 months (e.g., Food prices drop after harvest).
- **Residual:** The "Surprise" (War, Pandemic).



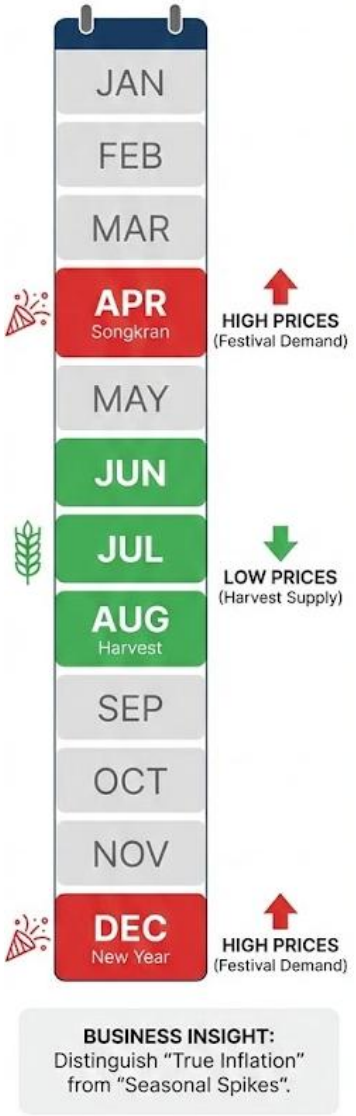
Implementing Decomposition in Python

- **Library:** statsmodels.tsa.seasonal
- **Function:** seasonal_decompose(data, model='additive', period=12)
- **Output:** We get four plots.
  - **Observed:** The raw data.
  - **Trend:** Smooth line.
  - **Seasonal:** Perfect waves.
  - **Resid:** Random noise + Shocks.



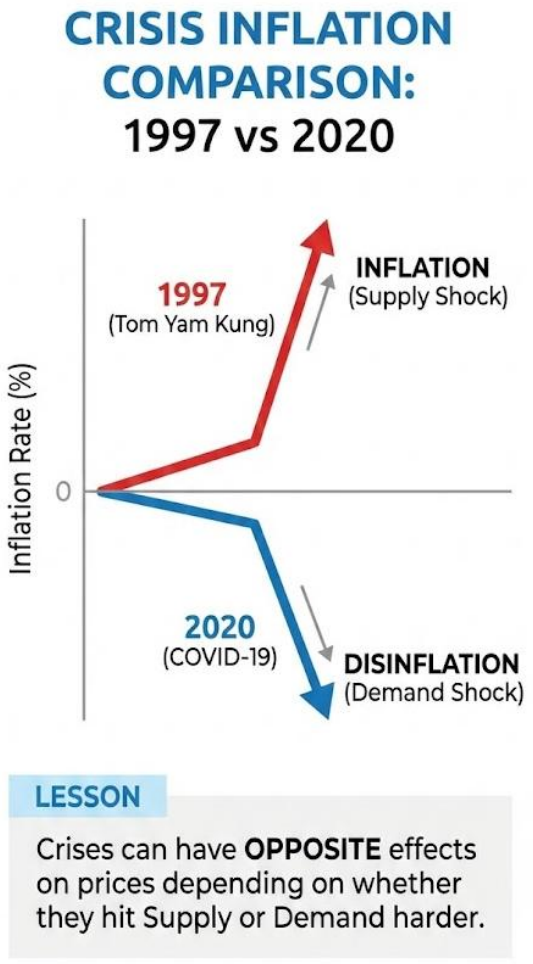
Analyzing the Seasonal Component

- Why do food prices oscillate?
- **Supply Side:** Harvest seasons increase supply (Price ↓).
- **Demand Side:** Festivals (e.g., Songkran, New Year, KinJ) increase demand (Price ↑).
- **Importance:** Businesses must distinguish between "True Inflation" and "Seasonal Spikes."



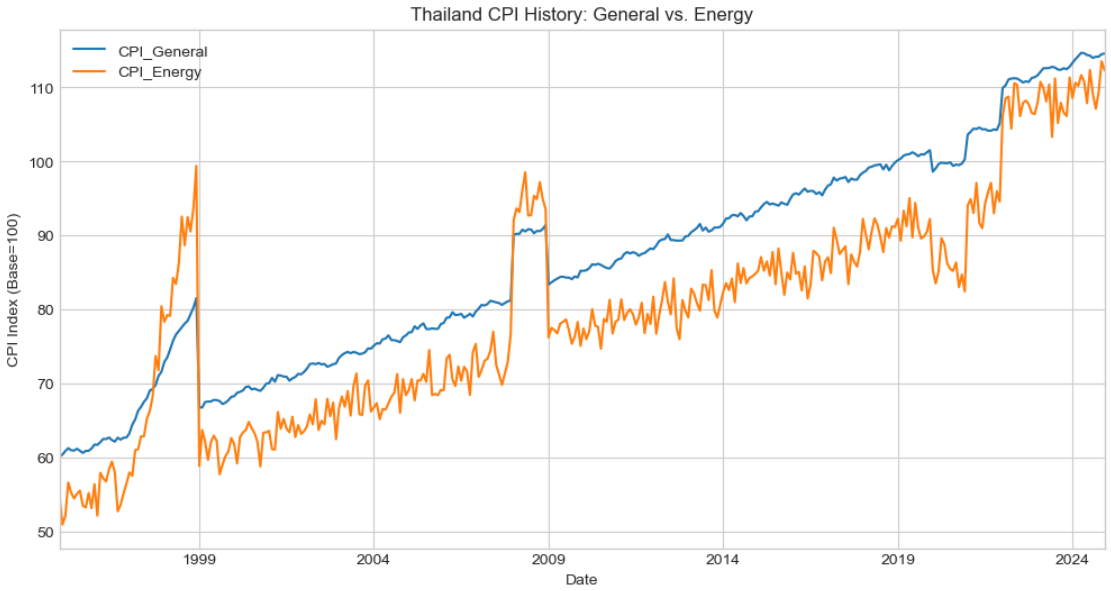
Visualizing the Crises (1997 vs 2020)

- We code a comparison plot of the Inflation Rate during the two crises.
- **1997 Line:** Shoots upwards (Inflation). Currency collapse made imports expensive.
- **2020 Line:** Dips downwards (Disinflation/Deflation). Lockdowns killed demand.
- **Lesson:** Crises can have opposite effects on prices depending on whether they hit Supply or Demand harder.



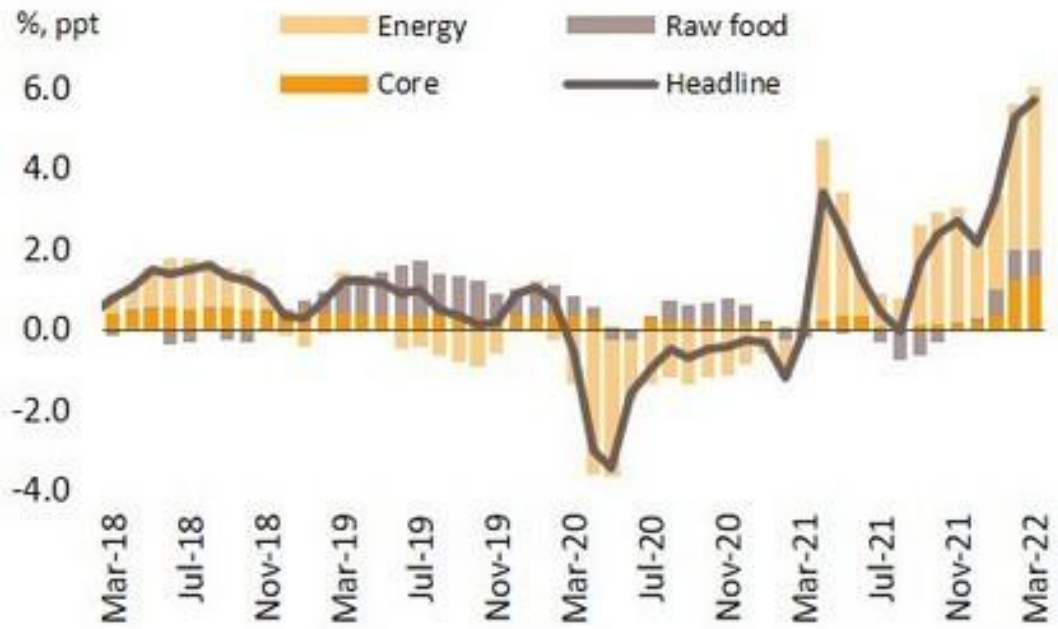
Volatility Analysis

- Not all prices move together.
- **Core Inflation:** Excludes Food and Energy. Why?
- **Answer:** Food and Energy are volatile (subject to weather/war).
- **In the Notebook:** We generate CPI_Energy with higher variance to simulate this volatility.



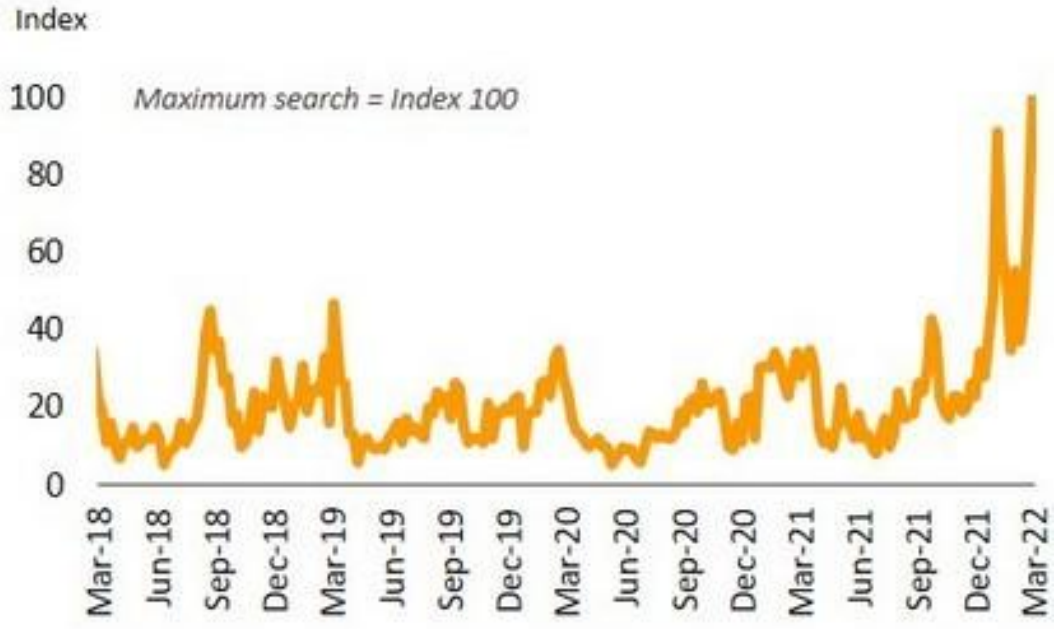
Thailand’s inflation from different sectors

Figure 1: Thai inflation



Source: Ministry of Commerce (MOC), Krungsri Research

Figure 2: Google search of the word “Inflation” in Thai

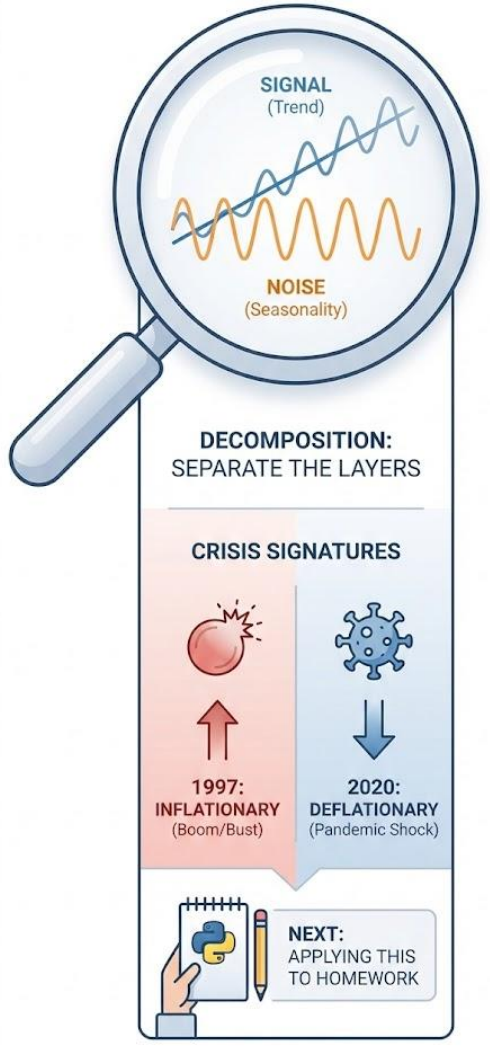


Source: Google trends, Krungsri Research

Headline inflation measures the total, raw inflation of an entire economy, including volatile food and energy costs, as reported in the Consumer Price Index (CPI). Core inflation excludes these volatile items to provide a more stable, underlying trend, making it the preferred metric for central banks to gauge long-term inflation trends.

Part 3 Summary

- **Personal CPI:** Depends on your specific basket.
- **Decomposition:** Helps separate signal (Trend) from noise (Seasonality).
- **Crisis signatures:** 1997 was inflationary; 2020 was deflationary.
- **Next:** Applying this to your homework.



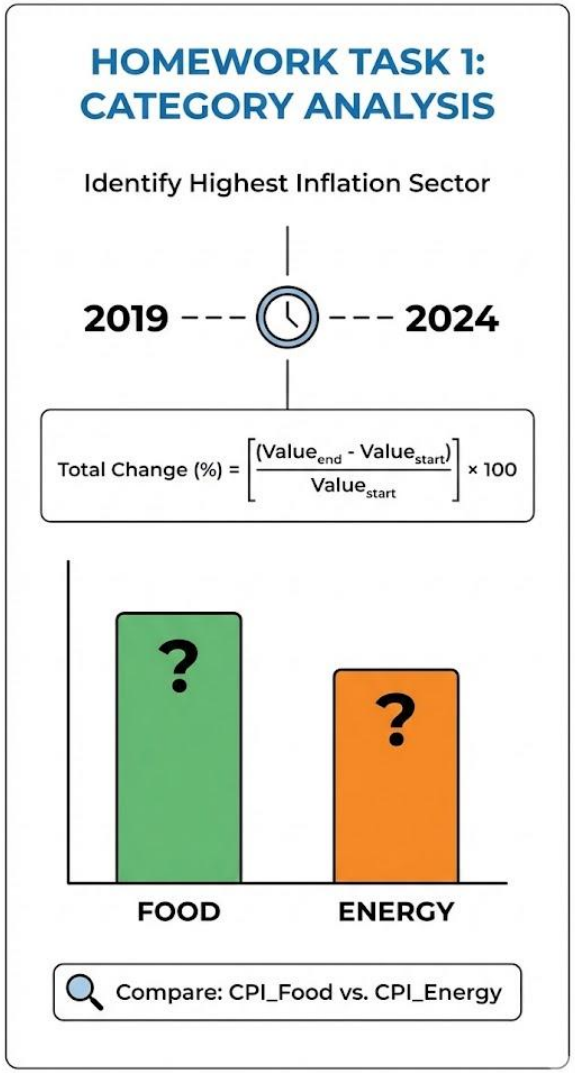
Part 4: Homework & Conclusion

Homework Task 1: Category Analysis

- **Objective:** Identify the highest inflation sector.
- **Data:** Filter the dataset for the last 5 years (2019-2024).
- **Calculation:**

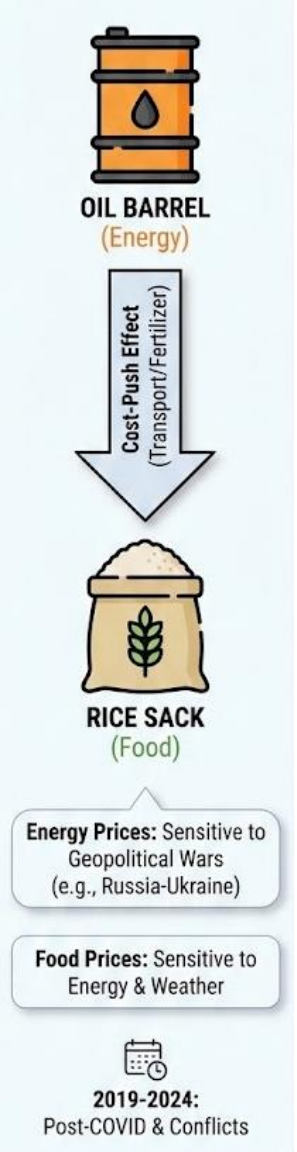
$$Total\ Change = \frac{\text{Value}_{end} - \text{Value}_{start}}{\text{Value}_{start}} \times 100$$

- **Comparison:** Compare CPI_Food vs. CPI_Energy.



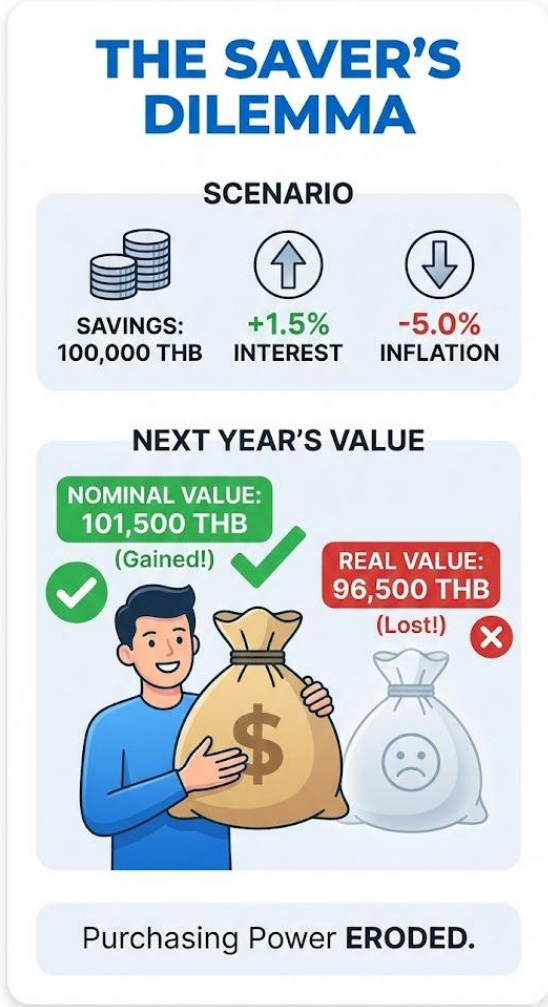
Hint for Task 1

- **Context:** The period 2019-2024 includes the post-COVID recovery and geopolitical conflicts (e.g., Russia-Ukraine).
- **Expectation:** Energy prices are highly sensitive to geopolitical wars (Cost-Push). Food prices are sensitive to both Energy (transport/fertilizer) and Weather.



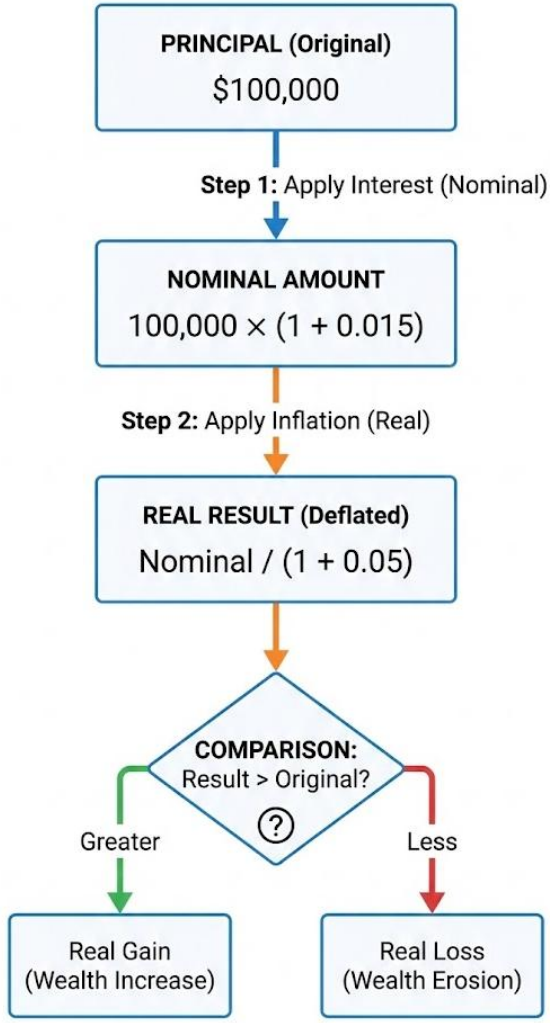
Homework Task 2: The Saver's Dilemma

- **Scenario:**
  - Savings: 100,000 THB.
  - Interest Rate: 1.5%.
  - Inflation Rate: 5.0%.
- **Task:** Calculate the **Nominal Value** next year and the **Real Value** next year.
- **Goal:** Prove whether you gained or lost purchasing power.



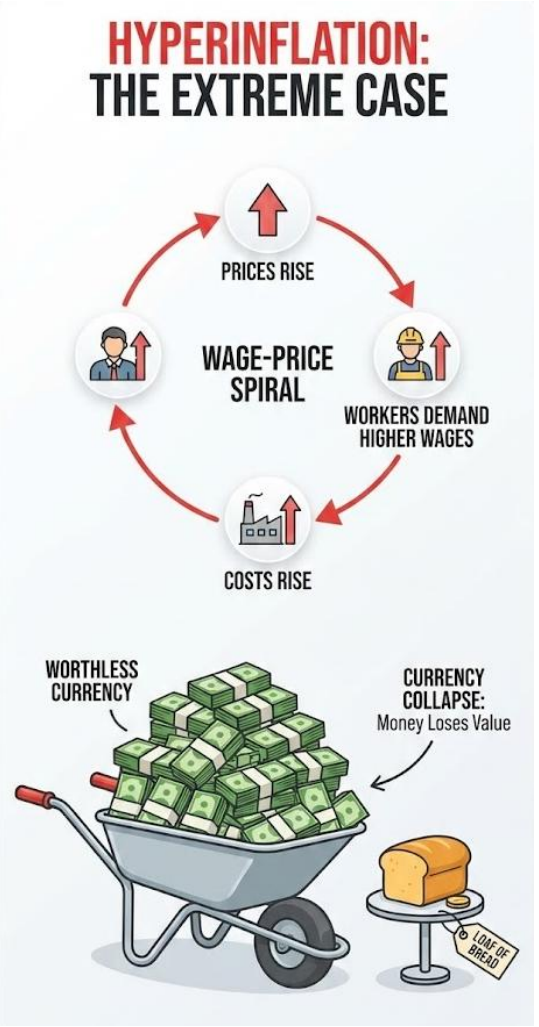
Hint for Task 2 (The Math)

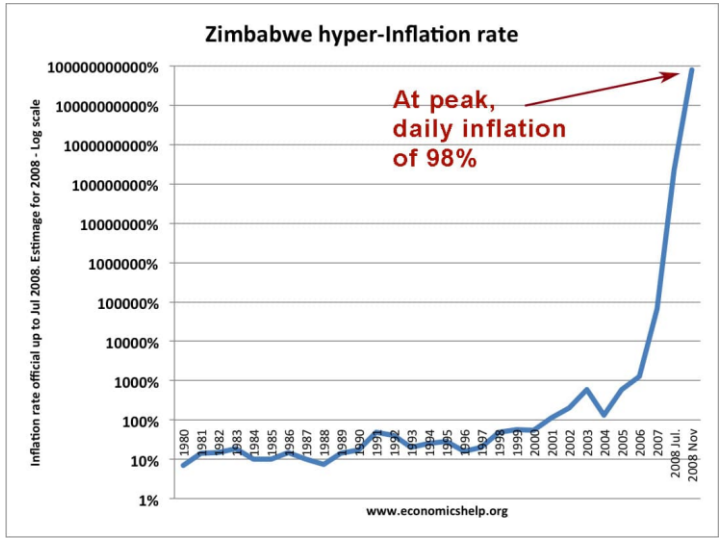
- **Step 1 (Nominal):** Multiply principal by (1 + Interest Rate).
  - 100,000 × 1.015
- **Step 2 (Real):** Deflate the Nominal amount by (1 + Inflation Rate).
  - Nominal/1.05
- **Comparison:** Is the result greater or less than the original 100,000?

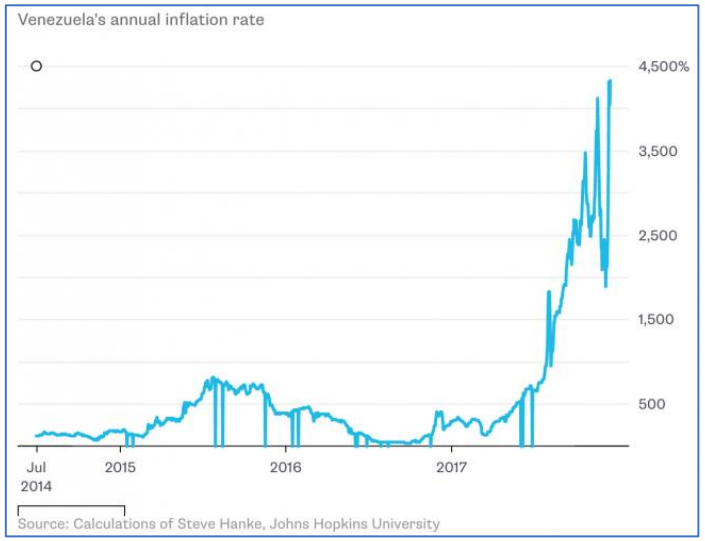


Hyperinflation (The Extreme Case)

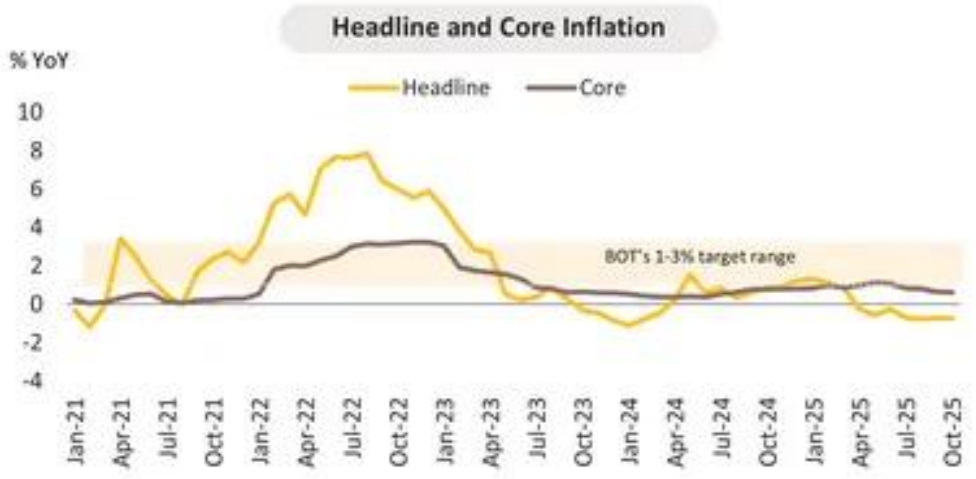
- What happens if inflation spirals out of control? (e.g., Zimbabwe, Venezuela).
- **Wage-Price Spiral:** Prices rise → Workers demand higher wages → Costs rise → Prices rise again.
- **Currency Collapse:** Money loses its function as a store of value. People resort to barter.

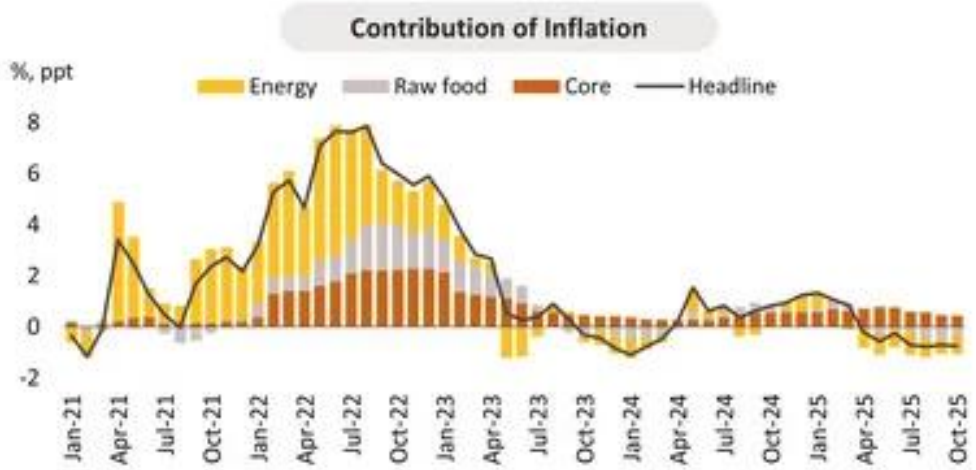




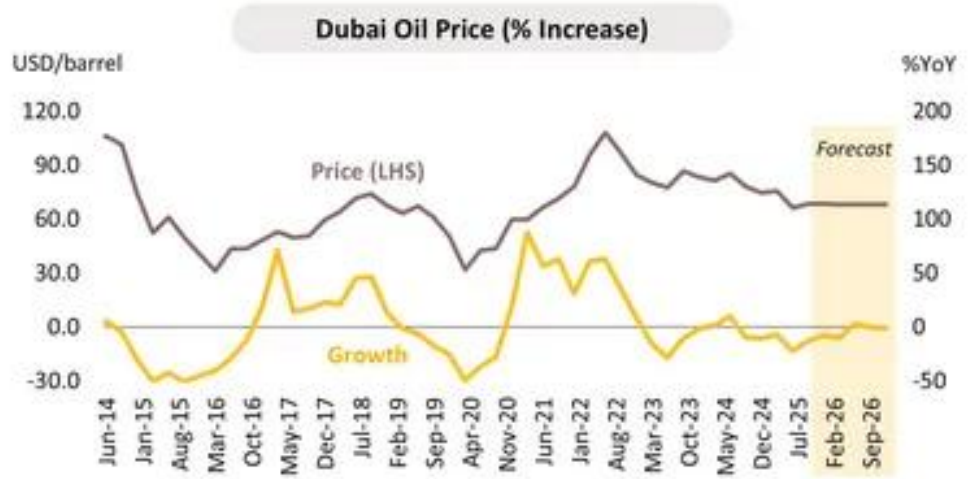


Thailand Inflation





Source: MOC, CEIC, Krungsri Research



Krungsri Research's view

- **Headline inflation is expected to remain low in 2026,** gradually returning to positive territory around 2Q26.
- Inflationary pressures will stay muted due to several factors. (i) Dubai crude oil prices are projected to remain broadly stable compared with late 2025, limiting cost-push pressures from the energy sector. (ii) Ongoing government cost-of-living measures, particularly energy-related subsidies, will continue to cap increases in electricity and fuel prices. (iii) Domestic demand is likely to recover only gradually, weighed down by high household debt, soft labor-market conditions in tourism-related sectors, and lingering political uncertainty in an election year—all of which will restrain demand-pull inflation.
- **Overall, we forecast headline inflation at 0.4% in 2026, up from –0.2% in 2025, but still well below the BOT's 1-3% target range.**


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Inflation Expectations

- **Self-Fulfilling Prophecy:** If everyone *thinks* inflation will be high next year, they buy now. This surge in demand *causes* high inflation.
- **Central Bank Role:** Managing *expectations* is as important as managing interest rates.



RESEARCH ARTICLE

Forecasting Inflation Expectations with Narrative Economics: Integrating Global Sentiment and Local Topic Modeling from Online News

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Abstract

Well-anchored inflation expectations are a cornerstone of effective monetary policy. However, traditional inflation-expectation measures that central bank monitors often suffer from data latency, failing to capture real-time shifts in public sentiment during crises or volatile inflation episodes. This study presents INFLEXION (Intelligent Framework for forecasting Inflation Expectations Incorporating Economic Sentiments and topics from ONline news), an automated expert system designed to nowcast and forecast Thailand's inflation expectations. We propose a novel multi-modal feature extraction pipeline that synthesizes topical signals, mined via Latent Dirichlet Allocation (LDA) from a corpus of 1.1 million news articles, with global sentiment scores from the GDELT project. Validated against official Bank of Thailand and the Ministry of Commerce survey data during the volatile post-pandemic period (2021–2023), our system demonstrates a significant divergence in forecastability between economic agents. The inclusion of news-based signals reduces prediction errors (RMSE) for **Household Inflation Expectations** by up to 46.84% on average during the COVID-19 pandemic phase in 2021 and 10.90% on average across the 2021-2023 period, compared to autoregressive baselines, validating the theory that households are "rationally inattentive" and heavily influenced by media narratives. Conversely, **Firm Inflation Expectations** exhibit resilience to news signals, suggesting a reliance on internal operational data. These findings establish INFLEXION as a viable decision-support tool for policymakers to monitor the de-anchoring of public expectations in real time.

- Under Review -

Final Recap

- **Business Cycles:** The economy breathes (Expansion/Recession).
- **Inflation:** The erosion of purchasing power.
- **CPI:** The standard metric using a Basket of Goods.
- **Real vs. Nominal:** Always adjust for inflation to see the truth.
- **Python:** Helps us visualize trends, seasonality, and shocks.



Course Roadmap

- **Next Session:** We look at the "Manager" of the economy.
- **Session 9:** Fiscal Policy & The Role of Government.
- Taxes, Budgets, and Public Goods.



Let’s check out our Notebook

